

Executive Summary

Delivery Performance & Customer Satisfaction | Olist Brazilian E-Commerce Marketplace

Purpose. This report looks at where delivery delays happen, how delivery time is related to customer reviews, and whether late orders can be identified early.

99,441 orders	R\$15.84M total revenue	12.5 days average lag	-0.33 lag / review	51% late orders caught
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Key Findings

1. Delays are concentrated in a few states.

The national average delivery lag is 12.5 days. The worst-performing states are RR (29.0), AP (26.7), AM (26.0), AL (24.0), and PA (23.3 days). These are mostly in or near Brazil's remote North region.

About 5.2% of orders (5,025) took more than 28.5 days, showing that a smaller group of orders has very long delays.

So what? Olist should focus improvement efforts on these areas instead of treating every region the same.

2. Slow delivery is linked to lower reviews.

Delivery lag and review score have a negative correlation of -0.33. Slower deliveries are generally connected with lower review scores. Delivery is not the only reason for a bad review, but it is an important part of the customer experience.

So what? Improving delivery will not guarantee 5-star reviews, but it can help reduce low scores.

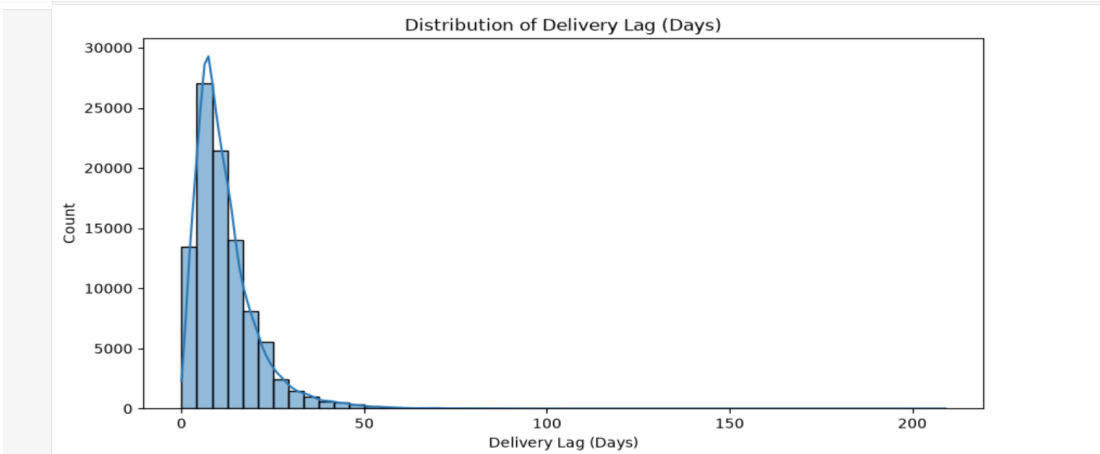
3. We can predict risky orders early.

A Logistic Regression model uses information available before shipment, including payment value, freight cost, order size, and customer state. It identified 51% of orders that eventually arrived late.

So what? Risky orders can be flagged early for a simple check, customer update, or carrier priority.

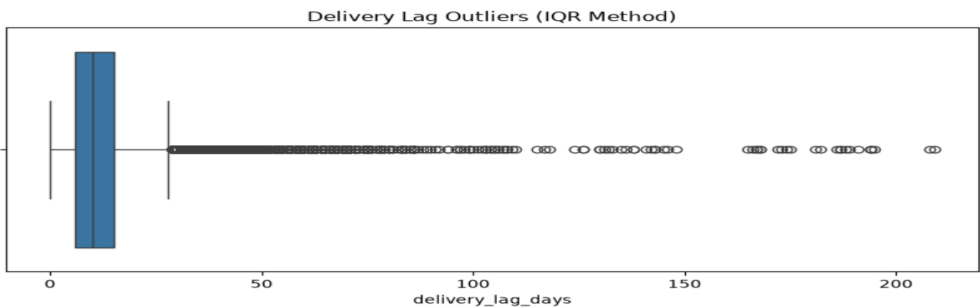
Delivery Lag Distribution

Most orders are delivered within the first few weeks, but a long tail of orders takes much longer.



Delivery Lag Outliers

The boxplot shows many observations beyond the normal range, supporting the finding that a smaller group of orders experiences very long delivery times.



Business Recommendations

The results point to a few practical actions. These recommendations are based directly on the analysis.

1. Focus on the highest-delay states

Work with local logistics partners or consider a distribution hub serving RR, AP, AM, AL, and PA. These states have the longest average delivery times.

Why: This targets the areas where the delivery problem is largest instead of spending the same effort across all regions.

2. Use the model as an early warning

The Logistic Regression model uses information available before shipment and identified 51% of orders that eventually arrived late.

What to do: Flag orders above the 50% risk threshold for a simple check, customer update, or carrier priority.

3. Track delivery lag regularly

Add delivery lag to the Power BI dashboard as a metric that managers check regularly. A simple alert could be used when a state's average lag goes above about 20 days.

Why: This can help the company notice a growing problem before it affects a large number of customers.

4. Improve small data-quality issues

The analysis found 4 zero-weight products and 8 orders marked as delivered without a delivery date. These issues should be corrected in the source data.

Why: Cleaner data makes future reports and prediction models more reliable.

Limitations

The dataset covers September 2016 to October 2018, so shipping conditions may have changed. The model is designed to catch more late orders, which means it also creates many false alarms. Around 9 out of 10 orders flagged as risky still arrive on time, so it should mainly be used as a low-cost warning or triage tool. Also, the -0.33 correlation shows a relationship, not proof that delivery time alone causes low reviews.

Conclusion

The main problem is not that every Olist delivery is slow. The biggest delays are concentrated in a few regions, especially the North. The most useful next steps are to focus on those regions, use the prediction model to flag risky orders, and monitor delivery lag regularly. These actions are simple and directly connected to the results of the analysis.